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## 1. Data Contract

### 1.1 QuerySpec

|                              |                                                           |
|------------------------------|-----------------------------------------------------------|
| query_text                   | string                                                    |
| query_profile                | GENERAL   RECENT   FOUNDATIONAL   EMERGING   RELATIONSHIP |
| keywords                     | list[string]                                              |
| topic_ids                    | list[string]                                              |
| researcher_ids               | list[string] or null                                      |
| institution_ids              | list[string] or null                                      |
| publication_types            | list[string] or null                                      |
| date_from, date_to           | date or null                                              |
| requested_relationship_types | list[string] or null                                      |
| broad_query                  | boolean                                                   |

The LLM/query parser may select only a predefined **query\_profile**. It must not generate arbitrary ranking weights.

### 1.2 CandidatePaper

#### Identity / metadata

- paper\_id, doi, openalex\_id, publication\_date, publication\_type, status
- author\_ids, venue\_id, topic\_ids

#### Retrieval

- bm25\_rank or null
- dense\_rank or null

#### GraphRAG

- bounded typed graph paths, hop counts, relationship types
- edge evidence confidences, best supported path, graph truncation flag

#### Evidence / provenance

- evidence confidence  $E(p) \in [0, 1]$
- evidence IDs, provenance records, source/version timestamps

#### Precomputed signals

- venue prior  $V'(p) \in [0, 1]$
- document prior  $D(p) \in [0, 1]$
- quality confidence  $c_Q \in [0, 1]$
- citation influence  $I(p) \in [0, 1]$
- innovation  $N(p) \in [0, 1]$

#### Author data

- query-conditioned authority  $A(p, q) \in [0, 1]$ , or deterministic historical author statistics sufficient to compute it

### 1.3 RankedPaper Output

Each eligible result must expose:

- paper\_id, final score  $S_z(p, q)$
- $H(p, q)$  hybrid retrieval relevance
- $G(p, q)$  graph relevance
- $R_z(p, q)$  combined relevance
- $Q(p)$  quality prior,  $I(p)$  citation influence,  $T_z(p)$  temporal validity
- $N(p)$  innovation,  $Q(p)N(p)$  quality-innovation interaction
- $A(p, q)$  author authority
- score breakdown, eligibility state, rejection reason if any
- winning graph path, evidence IDs, feature versions, ranking-config version
- deterministic tie-break key

If no result passes all gates, return:

**status = insufficient\_information.**

## 2. Mathematical Model

All primary component scores are normalized to  $[0, 1]$ .

## 2.1 Hybrid Retrieval Relevance

For retrieval systems  $m \in \{\text{BM25}, \text{Dense}\}$ , let  $r_m(p, q)$  be the candidate rank and let  $b_m$  be fixed system weights satisfying  $\sum_m b_m = 1$ .

$$H(p, q) = \frac{\sum_m \frac{b_m}{k_0 + r_m(p, q)}}{\sum_m \frac{b_m}{k_0 + 1}} \quad (1)$$

Initial setting:  $k_0 = 60$ . A missing candidate contributes zero for that retrieval system. BM25 and dense similarity must not appear again as independent final features after fusion.

## 2.2 Graph Relevance

For each valid graph path  $\ell$ :

$$g_\ell = c_\ell t_\ell \exp[-\rho(h_\ell - 1)] \quad (2)$$

where  $c_\ell$  is the minimum edge provenance confidence,  $t_\ell$  is query-conditioned relationship-type compatibility, and  $h_\ell$  is hop count.

$$G(p, q) = \max_\ell g_\ell \quad (3)$$

Use the best supported path rather than path count. Retain the winning path for explanation and debugging.

## 2.3 Combined Query Relevance

For query profile  $z$ :

$$R_z(p, q) = (1 - \gamma_z)H(p, q) + \gamma_z G(p, q) \quad (4)$$

For non-relationship queries, enforce a minimum textual relevance  $H(p, q)$  so graph connectivity cannot rescue a textually unrelated paper.

## 2.4 Conservative Quality Prior

Shrink a field- and venue-type-normalized venue statistic  $V(p)$  toward neutral:

$$V'(p) = \frac{n_v V(p) + \kappa(0.5)}{n_v + \kappa} \quad (5)$$

Then define:

$$Q(p) = c_Q [0.6V'(p) + 0.4D(p)] + (1 - c_Q)0.5 \quad (6)$$

$Q$  is a conservative prior, not an objective methodological-quality score. Do not include paper citations, author prestige, or evidence completeness in  $Q$ . Retraction or invalid publication status causes rejection.

## 2.5 Citation Influence

Preferred MVP:

$$I(p) = \text{field/year/document-type normalized citation percentile} \quad (7)$$

If computed locally, use weakly source-weighted citations:

$$C_w(p) = \sum_{j:j \rightarrow p} \text{clip}(0.5 + 0.5V(j), 0.5, 1) \quad (8)$$

and cohort-normalize:

$$I(p) = F_{f,y,d}(\log(1 + C_w(p))) \quad (9)$$

For sparse cohorts, back off to the parent field and shrink toward 0.5. Never use raw lifetime citation count directly.

## 2.6 Query-Conditioned Author Authority

Using relevant *prior* papers only:

$$A(u, q) = 0.60P_{\text{impact}}(u, q) + 0.25P_{\text{output}}(u, q) + 0.15P_{\text{activity}}(u, q) \quad (10)$$

where:

- $P_{\text{impact}}$ : percentile of topic-weighted normalized prior influence;
- $P_{\text{output}}$ : percentile of  $\log(1 + \text{relevant prior publications})$ ;
- $P_{\text{activity}}$ : recent relevant-topic activity.

Exclude the candidate paper itself. For multiple authors use a trimmed mean or top-two mean, never maximum prestige. For paper ranking enforce  $w_A \leq 0.05$ .

## 2.7 Innovation

Construct *EarlierComparable*( $p$ ) from papers published before  $p$  that share a sufficiently confident field/topic and have usable title/abstract embeddings. Require at least 20–50 comparable papers and minimum contextual similarity. Use an external reference corpus if the local corpus is sparse.

For initial  $k = 5$  nearest earlier comparable papers:

$$d_{\text{sem}}(p) = 1 - \frac{1}{k} \sum_{d \in NN_k(p)} \cos(e_p, e_d) \quad (11)$$

Convert the distance to a field/year novelty percentile:

$$P_{\text{sem}}(p) = F_{f,y}^{\text{novelty}}(d_{\text{sem}}(p)) \quad (12)$$

Then:

$$N(p) = c_N \max\left(0, \frac{P_{\text{sem}}(p) - 0.5}{0.5}\right) \quad (13)$$

$c_N$  reflects abstract availability, topic confidence, and comparison-set size. If novelty is unreliable, set  $N(p) = 0$ . Innovation contributes only through  $Q(p)N(p)$ .

## 2.8 Temporal Validity

Let  $a(p)$  be paper age in years,  $h_z$  the profile half-life, and  $\beta_z$  the innovation-based half-life extension coefficient:

$$h_{\text{eff}}(p, z) = h_z [1 + \beta_z Q(p)N(p)] \quad (14)$$

$$T_z(p) = 2^{-a(p)/h_{\text{eff}}(p, z)} \quad (15)$$

Constraint:  $0 \leq \beta_z \leq 0.5$ . For RECENT, set  $\beta_z = 0$ .

## 2.9 Eligibility Gates

Apply structured filters first. A candidate is eligible only if:

$$R_z(p, q) \geq \tau_R, \quad E(p) \geq \tau_E, \quad \text{status}(p) = \text{valid} \quad (16)$$

For general/non-relationship queries also require:

$$H(p, q) \geq \tau_H \quad (17)$$

Initial hypotheses:  $\tau_R = 0.30$ ,  $\tau_E = 0.70$ .

## 2.10 Final Paper Score

For eligible paper  $p$  under profile  $z$ :

$$S_z(p, q) = w_R R_z(p, q) + w_Q Q(p) + w_I I(p) + w_T T_z(p) + w_N Q(p)N(p) + w_A A(p, q) \quad (18)$$

with:

$$w_R + w_Q + w_I + w_T + w_N + w_A = 1 \quad (19)$$

Final acceptance requires  $S_z(p, q) \geq \tau_S$ . Initial hypothesis:  $\tau_S = 0.45$ .

### 2.11 Topic Diversification

Diversity is list-level, not a paper feature. For already-selected set  $L$ :

$$\text{MMR}(p) = \eta S_z(p, q) - (1 - \eta) \max_{d \in L} \text{sim}_{\text{topic}}(p, d) \quad (20)$$

Initial settings: broad exploratory  $\eta = 0.80$ ; moderately broad  $\eta = 0.90$ ; narrow/fact-seeking: skip MMR or use  $\eta \geq 0.95$ .

Required order: deduplicate  $\rightarrow$  gates  $\rightarrow$  base score  $\rightarrow$  strong candidate pool  $\rightarrow$  optional MMR  $\rightarrow$  Top-K. MMR must never admit candidates below the relevance floor.

### 2.12 Researcher Ranking

Calculate eligible paper scores with the paper-level author prior disabled. For the researcher’s Top- $L$  papers, use:

$$d_j = \frac{1}{\log_2(j + 1)} \quad (21)$$

$$P(u, q) = \frac{\sum_{j=1}^L d_j S(p_j, q)}{\sum_{j=1}^L d_j} \quad (22)$$

Initial researcher score:

$$S_{\text{researcher}}(u, q) = 0.70P(u, q) + 0.20A(u, q) + 0.10G_u(u, q) \quad (23)$$

For RELATIONSHIP queries, transfer weight from  $A$  to  $G_u$ . Do not use raw publication count, h-index, total career citations, or maximum single-paper score as the primary researcher ranking.

## 3. Feature Ownership

| Owner                       | Features / Responsibilities                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database / Data Engineering | Canonical researcher/work/institution/venue/topic IDs; DOI/OpenAlex/ORCID/ROR aliases; titles, abstracts, publication dates/types; authorship and affiliation intervals; embeddings and model version; topics, confidence and taxonomy version; citation edges and yearly counts; historical venue statistics; retraction/correction/version relations; evidence/provenance records; normalization cohort statistics. |
| Retrieval / GraphRAG        | BM25 rank; dense rank; candidate generation; bounded typed graph paths; hop counts; relationship types; edge confidence; best supported graph path; graph truncation metadata; candidate evidence IDs.                                                                                                                                                                                                                |
| Ranking                     | $H, G, R$ ; normalization/calibration; eligibility thresholds; fixed profile weights; $Q$ ; normalized $I; T; A; QN$ interaction; final score; researcher aggregation; deduplication; MMR; score breakdowns; deterministic tie-breaking; ranking/config versions.                                                                                                                                                     |
| Offline Feature Pipeline    | Citation percentile/normalized influence; author historical statistics; semantic innovation $N$ ; optional future novelty/network features.                                                                                                                                                                                                                                                                           |
| LLM / Query Parser          | May parse query, select one validated profile, extract keywords/topics/filters/relationships, and synthesize final answers from supplied evidence. Must not invent weights; assign $Q/I/A/N$ scores; determine provenance truth; manufacture graph edges; override filters/gates; or choose unsupported citations.                                                                                                    |

## 4. Query Profile Table

| Profile      | $w_R$ | $w_Q$ | $w_I$ | $w_T$ | $w_N$ | $w_A$ | $\gamma$ | Half-life |
|--------------|-------|-------|-------|-------|-------|-------|----------|-----------|
| GENERAL      | .55   | .16   | .12   | .08   | .06   | .03   | .10      | 8y        |
| RECENT       | .50   | .14   | .07   | .23   | .03   | .03   | .08      | 2y        |
| FOUNDATIONAL | .42   | .14   | .32   | .03   | .04   | .05   | .10      | 25y       |
| EMERGING     | .46   | .14   | .06   | .10   | .21   | .03   | .10      | 4y        |
| RELATIONSHIP | .63   | .12   | .08   | .06   | .08   | .03   | .35      | 10y       |

| Profile      | $\beta$ |
|--------------|---------|
| GENERAL      | .25     |
| RECENT       | .00     |
| FOUNDATIONAL | .50     |
| EMERGING     | .15     |
| RELATIONSHIP | .25     |

### Profile selection rules

- RECENT: recent, latest, current, newest
- FOUNDATIONAL: foundational, classic, seminal, influential
- EMERGING: novel, emerging, innovative, new direction
- RELATIONSHIP: collaboration, co-author, affiliation, citation path, relationship
- GENERAL: default

Explicit date constraints are always hard filters. Multiple intents require a predefined combined profile, never generated runtime weights.

## 5. Offline / Runtime Boundary

### 5.1 Offline / Periodic Precomputation

- identity reconciliation
- embeddings and topic assignment
- field/year/type citation cohorts
- venue historical statistics
- citation influence percentiles
- author prior-publication statistics
- provenance/evidence confidence
- retraction/correction status
- EarlierComparable sets and historical nearest neighbours
- novelty distances, percentiles, confidence and final  $N(p)$

Version offline features by source snapshot, embedding model, topic taxonomy, normalization configuration, and feature version.

### 5.2 Runtime Pipeline

User query

- > Query parser / LLM
- > validated QuerySpec
- > BM25 + dense retrieval
- > RRF candidate set
- > bounded GraphRAG expansion
- > candidate signals + evidence
- > structured filters
- > eligibility gates
- > deterministic ranking
- > deduplication
- > optional MMR
- > Top-K evidence
- > LLM synthesis
- > final answer + deterministic citations

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### 5.3 Runtime Ranking May

- calculate  $H$ ,  $G$ ,  $R$ ,  $T$ ,  $QN$ , and final  $S$
- read precomputed  $Q$ ,  $I$ ,  $N$
- calculate/read  $A$  from deterministic historical author statistics
- threshold, tie-break, deduplicate, and optionally apply MMR

### 5.4 Runtime Ranking Must Not Require Per-Candidate LLM Calls For

- quality estimation
- innovation estimation
- influence estimation
- author authority/prestige estimation
- provenance verification
- graph scoring
- weight generation

## 6. Evaluation Plan

### 6.1 Judged Dataset

Initial target: 50–100 manually judged queries spanning GENERAL, RECENT, FOUNDATIONAL, EMERGING, RELATIONSHIP, broad exploratory, narrow factual, weak-evidence, cross-discipline, and early-career cases.

Use graded judgments: *highly relevant*, *relevant*, *marginal*, *irrelevant*.

### 6.2 Metrics

|                        |                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Retrieval</b>       | Recall@50; verifies candidate generation before ranking.                                                                |
| <b>Ranking</b>         | Primary: nDCG@10. Secondary: MRR. Optional: Precision@K, Recall@K.                                                      |
| <b>Diversification</b> | Subtopic recall and $\alpha$ -nDCG for broad queries; compare base ranking vs. base + MMR.                              |
| <b>Evidence</b>        | Weak-evidence rejection rate, false acceptance, false rejection, and <code>insufficient_information</code> correctness. |
| <b>Profiles</b>        | Query-profile selection accuracy and ranking metrics per profile.                                                       |

### 6.3 Required Ablations

Evaluate with each component removed independently:

- graph relevance
- author authority
- venue/quality prior
- temporal score
- innovation interaction
- MMR

Mandatory fairness ablation:  $w_A = 0$ .

### 6.4 Robustness Slices

Evaluate separately on early-career researchers, established researchers, recent low-citation papers, old highly cited papers, cross-discipline papers, small/unknown venues, sparse metadata, weak topic confidence, and high-degree graph nodes.

### 6.5 Sanity Cases

- A. Famous author + irrelevant paper  $\rightarrow$  reject via relevance gate.
- B. Very recent + weak paper  $\rightarrow$  recency must not dominate.
- C. Old foundational + high influence  $\rightarrow$  may rank highly under FOUNDATIONAL.
- D. New + high  $Q$  + high  $N$  + few citations  $\rightarrow$  competitive under EMERGING.
- E. Very unusual + low  $Q$   $\rightarrow$  limited by  $QN$ .
- F. Old cited paper vs. strong recent paper  $\rightarrow$  ordering changes by profile.
- G. Weak provenance  $\rightarrow$  reject regardless of prestige.

### 6.6 Parameter Tuning

Tune with constrained grid search:

$$\{w_*, \tau_R, \tau_H, \tau_E, \tau_S, \gamma_z, h_z, \beta_z, \eta\}.$$

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Constraints:

- relevance remains dominant;
- $w_A \leq 0.05$  for paper ranking;
- profile count remains small;
- all parameters remain interpretable and versioned;
- no LLM-generated weights.

### 6.7 MVP Acceptance Criteria

1. Recall@50 is sufficient for candidate generation.
2. nDCG@10 improves over a pure RRF baseline.
3. RECENT and FOUNDATIONAL produce meaningfully different rankings.
4. Weak-evidence candidates are reliably rejected.
5. Setting  $w_A = 0$  does not cause major ranking collapse.
6. Innovation improves EMERGING queries without degrading general ranking.
7. MMR improves broad-query coverage without violating relevance gates.
8. Identical query, candidates, config and feature versions produce identical rankings.
9. Every result exposes a score breakdown and evidence IDs.
10. No runtime LLM call assigns ranking feature scores.